

Identifying Key Predictors of Cognitive Status Using Machine learning

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Problem

Predicting the incidence of dementia is crucial for making decisions, such as follow-ups. To better make them, prediction models have been developed. The accuracy of these models can be heavily reliant on both the quality and quantity of the data it is fed. Identifying the most important predictors for proper weighting is priority

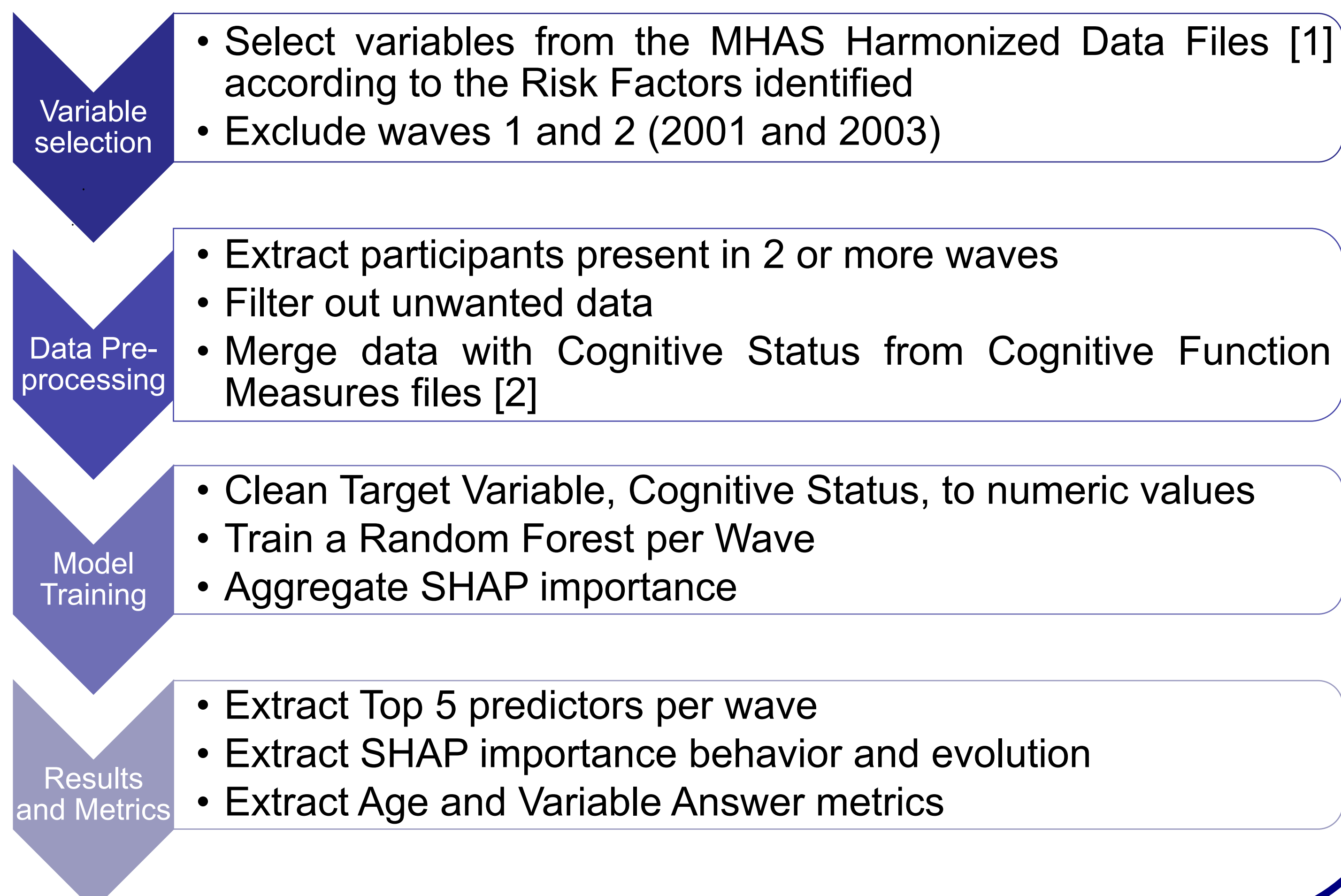
Objectives

Our objective is to identify the most important predictors for dementia prediction using a Machine Learning Algorithm by analyzing their relevance through time, thus ensuring the data's quality.

By doing this we can:

- Identify the top 5 most important predictors across the different waves of application
- Follow the evolution and change of this importance throughout the 4 waves of application

Methods



Results

Wave	Top 1	Top 2	Top 3	Top 4	Top 5
2012	Social Isolation	Time Use	Vision	Obesity	Hypertension
2015	Time Use	Social Isolation	Hospitalization	Hypertension	Vision
2018	Time Use	Social Isolation	Hypertension	Education	Falls
2021	Time Use	Social Isolation	Education	Falls	Hypertension

Table 1, Top 5 Predictors per Wave

After the analysis and the extraction of the metrics we see a pattern. On all 4 waves the top 2 predictors remained Social Isolation and Time Use.

Next is Hypertension, appearing 4 times, followed by Education, Falls and Vision, all appearing twice

Finally, both Obesity and Hospitalization appear only once, rising on their respective wave of appearance and then proceeding to fall again, maintaining a more constant shift in importance.

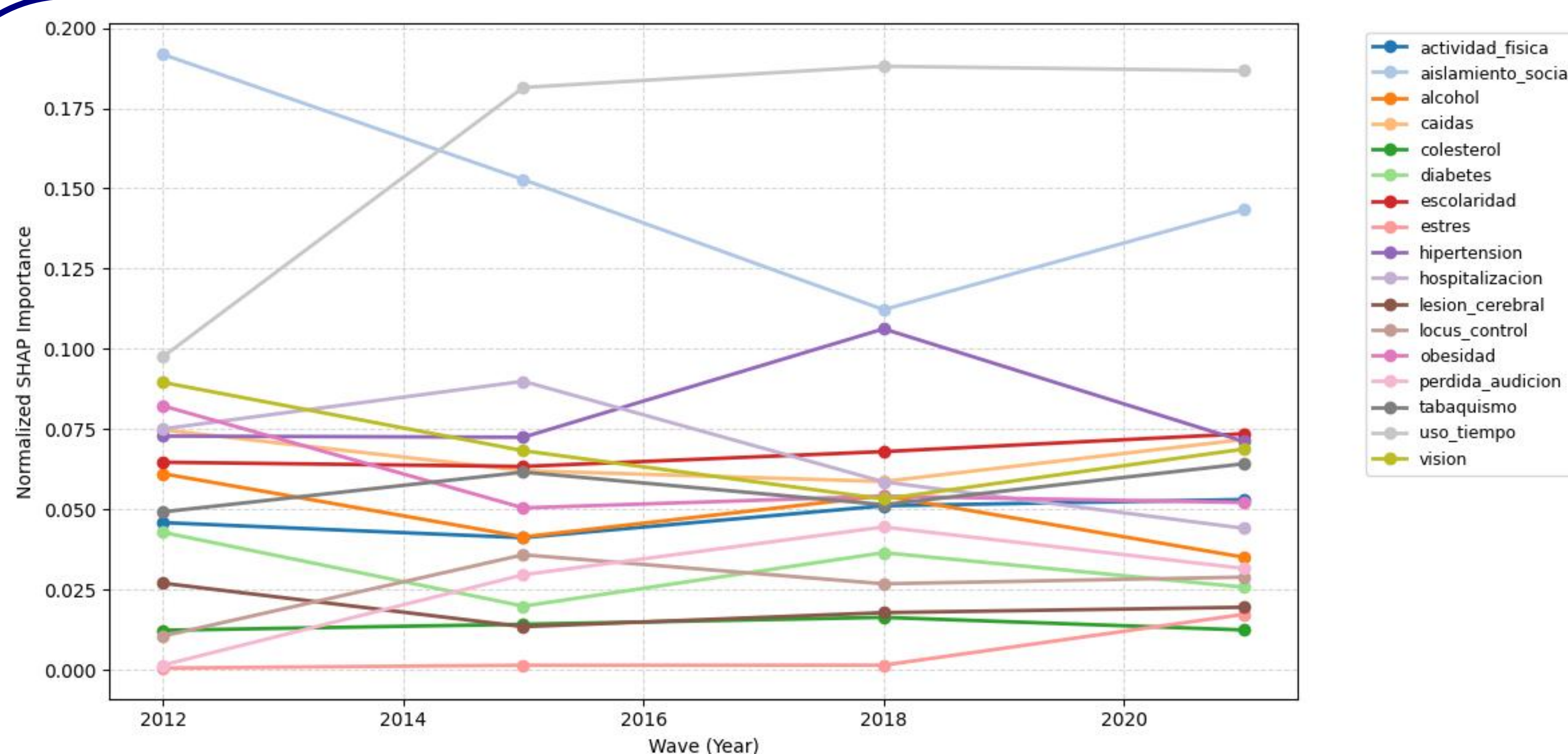


Figure 1, Predictor Importance Evolution

Conclusions

Based on the analysis results, the top 5 predictors as follows:

1. Use of Time
2. Social Isolation
3. Hypertension
4. Education
5. Falls

Future Work

Random Forests can be prone to both overfitting and biased predictions. A future measure to better reaffirm this ranking would be to train a RF model with a historical approach. This one would train on the dataset without separating waves.

References

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